PS200D: Regression Discontinuity Designs

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Roadmap

1. Theory: Potential outcomes, identification, key quantities
2. Randomization
   - difference in means, variance estimation
   - covariate adjustment
   - blocking
   - cluster randomization
3. Selection on Observables
   - sub-classification
   - matching (on X)
   - weighting (on X)
   - matching and weighting on $Pr(D = 1)$ (propensity scores)
   - regression
4. Difference in Differences and Synthetic Control
5. Instrumental Variables
6. Regression Discontinuity: Sharp, Fuzzy
7. Sensitivity and Bounds

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- Widely applicable in a rule based world (administrative programs, elections, etc.)
- High internal validity (see e.g. Cook, Shadish, Wong 2008)

Outline

1. Sharp Regression Discontinuity Design
   - Identification
   - Sharp RRD Examples
   - Sharp RDD: Estimation
   - Falsification Checks
2. Fuzzy Regression Discontinuity Design
   - Identification
   - Estimation
   - Fuzzy RDD Example: The Effect of Early Release from Prison Marie (2008)

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Sharp Regression Discontinuity Design

- Imagine a binary treatment D_i that is completely determined by the value of a predictor X_i being on either side of a fixed cutoff point c:

$$D_i = \mathbb{1}_{\{X_i > c\}} \text{ so } D_i = \begin{cases} D_i = 1 & \text{if } X_i > c \\ D_i = 0 & \text{if } X_i < c \end{cases}$$

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- Situation arises often from administrative decisions, where policy decisions are made using sharp rules rather than administrator discretion

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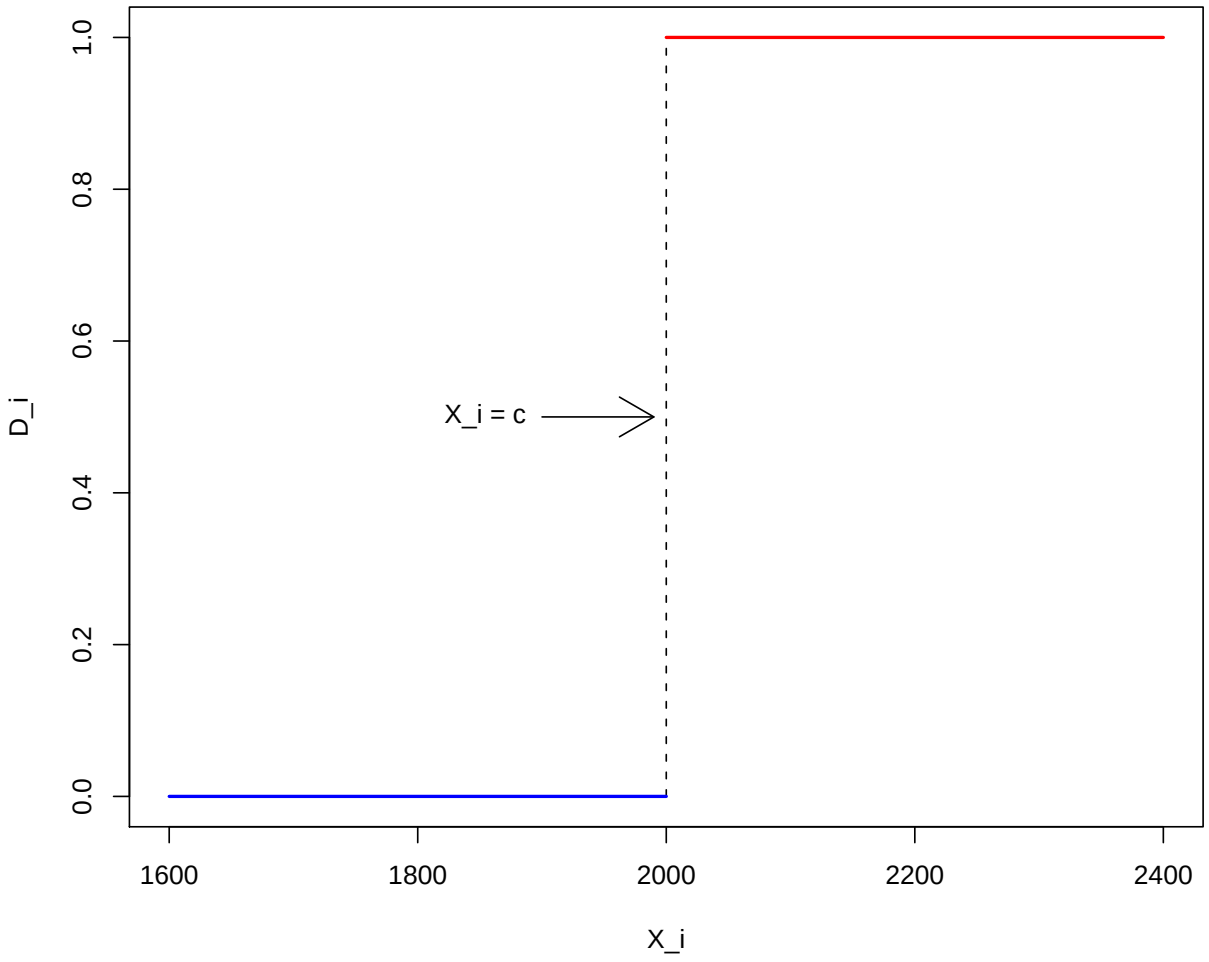
  - Y_{0i} denotes potential earnings without the scholarship
  - Y_{1i} denotes potential earnings with the scholarship
- Y_{1i} and Y_{0i} are correlated with X_i: on average, students with higher SAT scores obtain higher earnings

Sharp RDD: Graphical Illustration



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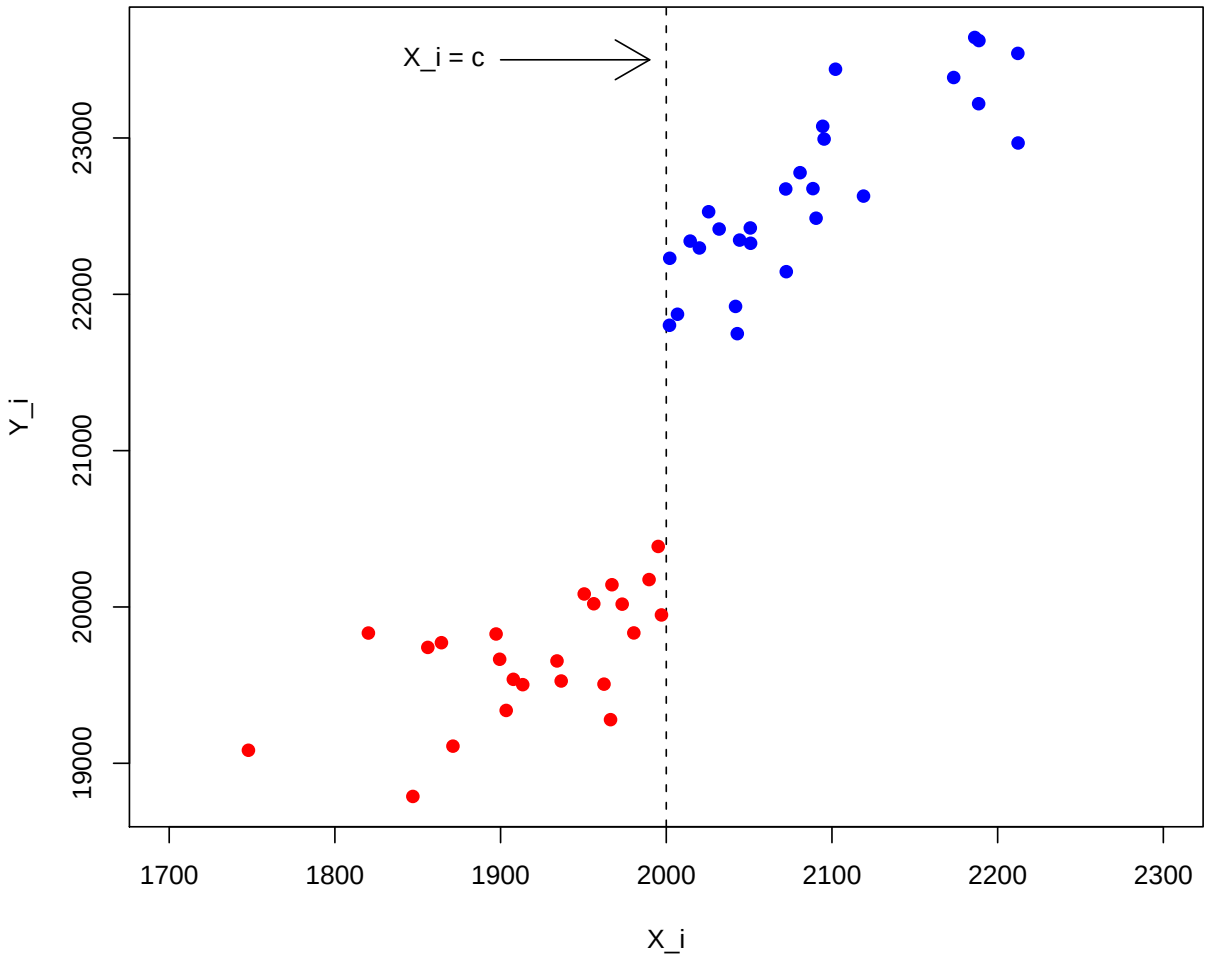

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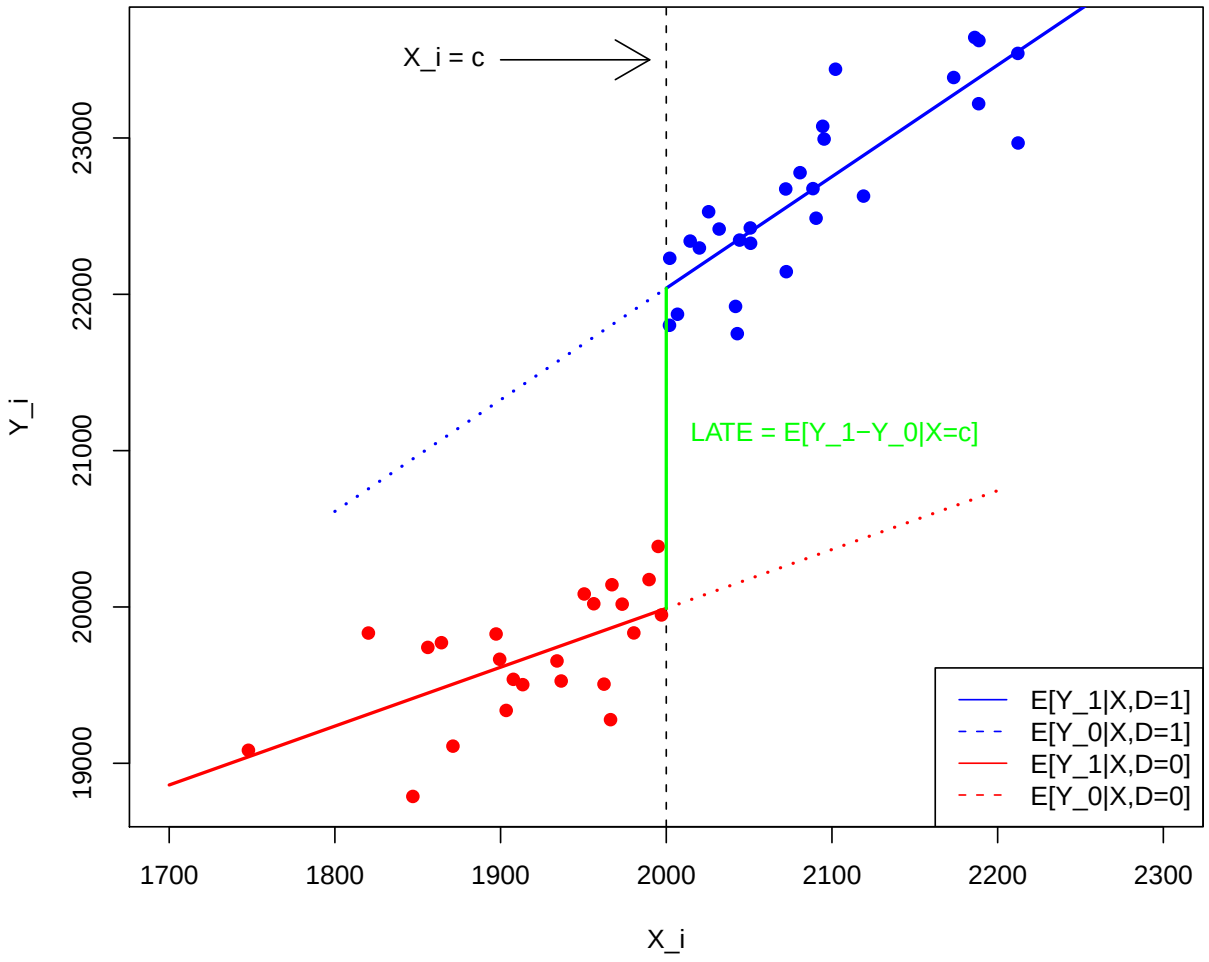

Sharp RDD: Identification

Identification Assumption

$\mathbb{E}[Y_{1i}|X_i]$ *and* $\mathbb{E}[Y_{0i}|X_i]$ *are continuous in* X_i *around the threshold* $X_i = c$

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Identification Assumption

$\mathbb{E}[Y_{1i}|X_i]$ and $\mathbb{E}[Y_{0i}|X_i]$ are continuous in X_i around the threshold $X_i = c$

Identification Result

The treatment effect is identified at the threshold as:

$$\begin{aligned}
\alpha_{SRDD} &= \mathbb{E}[Y_{1i} - Y_{0i}|X_i = c] \\
&= \mathbb{E}[Y_{1i}|X_i = c] - \mathbb{E}[Y_{i0}|X_i = c] \\
&= \lim_{x \downarrow c} \mathbb{E}[Y_{1i}|X_i = c] - \lim_{x \uparrow c} \mathbb{E}[Y_{i0}|X_i = c]
\end{aligned}$$

Without further assumptions α_{SRDD} is only valid at the threshold.

Comments on Identification

The continuity assumption says a lot:

- By saying $\mathbb{E}[Y_{1i}|X_i]$ and $\mathbb{E}[Y_{0i}|X_i]$ are continuous in X near c, we mean they are "smooth", or do not "jump" at $X = c$.
- This is why we can say:

$$\mathbb{E}[Y_{1i}|X_i = c] = \lim_{\epsilon \downarrow 0} \mathbb{E}[Y_{i1}|X_i = c + \epsilon, D_i = 1]$$

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- Implies any *unobserved influences on the potential outcomes* are also continuous in X (at $X = c$)
- You can also see it as our good old:

$$\{Y_{1i}, Y_{0i}\} \perp\!\!\!\perp D_i | X_i = c$$

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Duverger's Law (Eggers, 2010)

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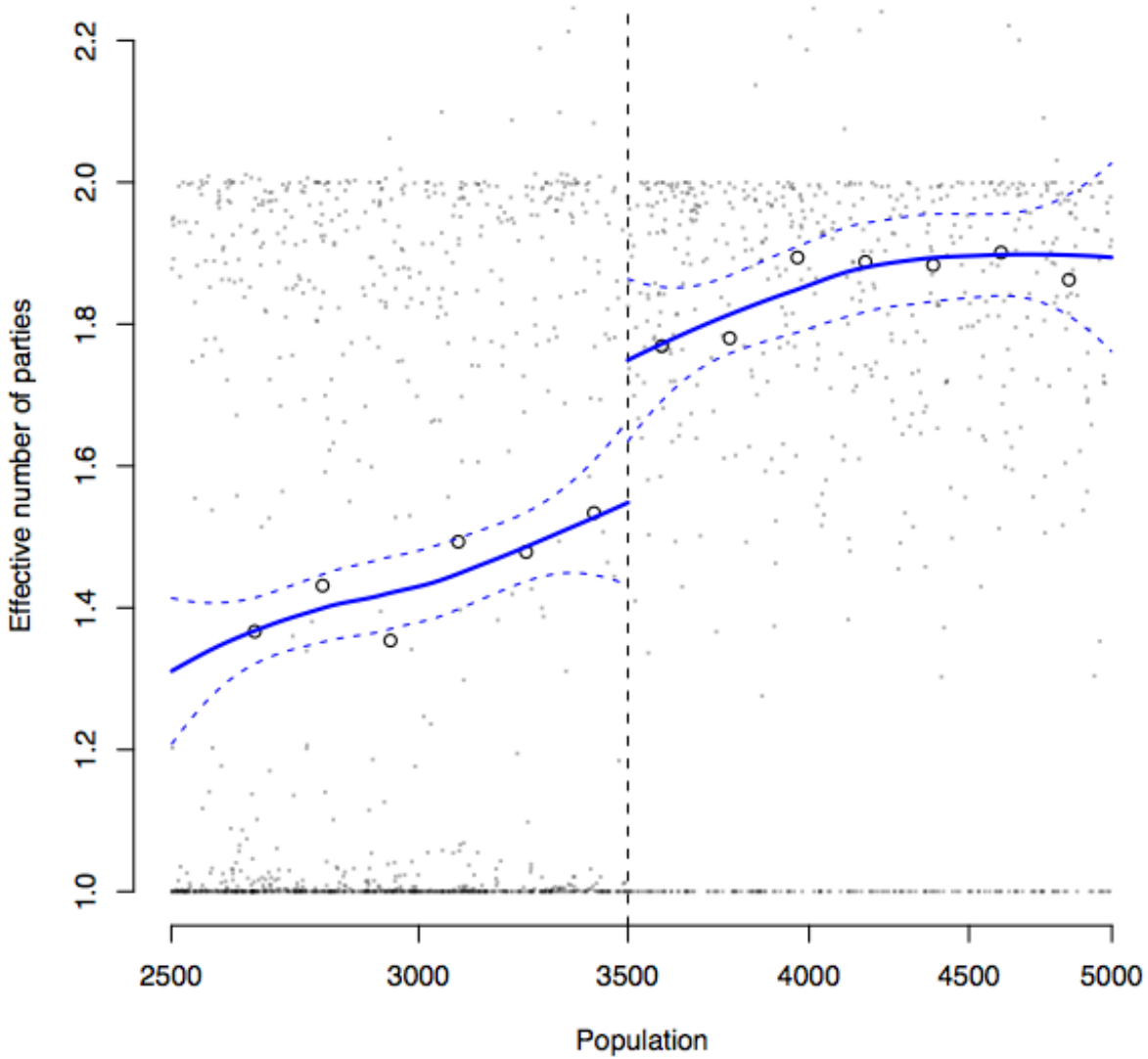
- In French municipalities, the electoral rule used to elect the municipal council depends on the city's population:
  - cities with fewer than 3,500 people use first-past-post
  - cities with a population of 3,500 or more use a form of PR rule

Sharp RDD: Duverger's Law



Sharp RDD: Electronic Voting (Hidalgo, 2012)

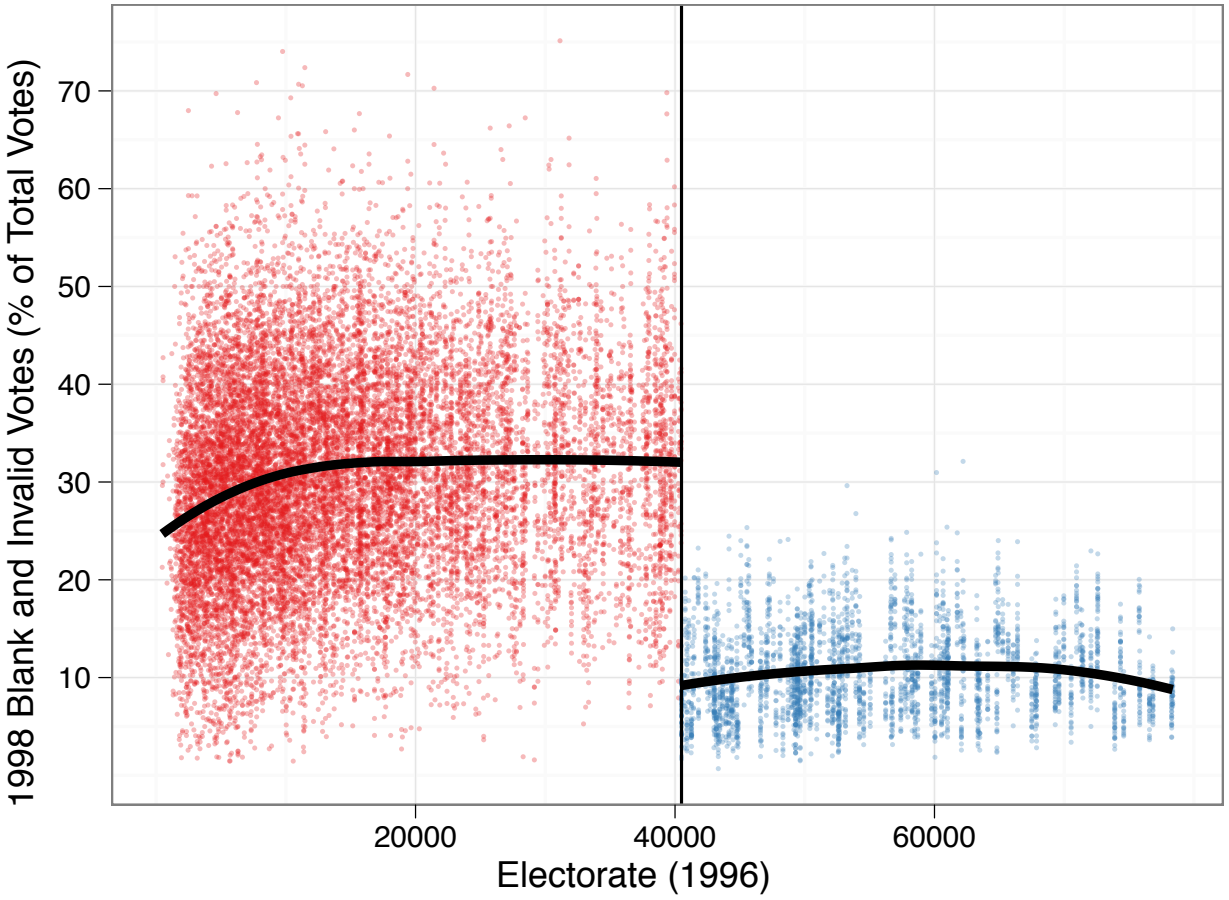
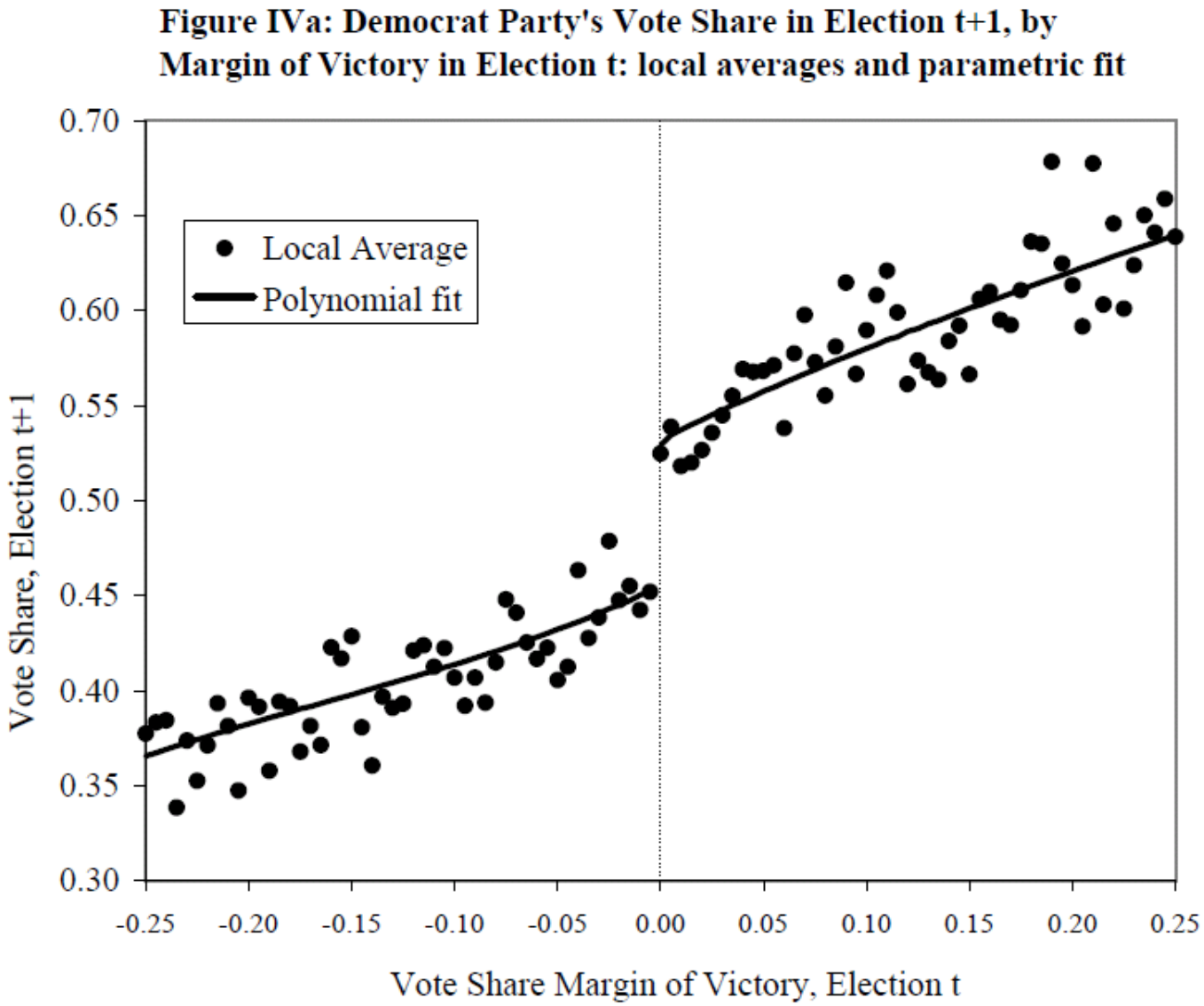


Figure 6: The effect of electronic voting on the percent of null and blank votes. Each dot is a polling station. Polling stations to the left of the vertical black line used paper ballots and polling stations to the right used electronic voting. The black horizontal line is the conditional mean of the outcome estimated with a loess regression.

Sharp RDD: Incumbency Advantage



Other Recent Examples

- Effect of class size on student achievement (Angrist and Lavy 1999)
- Effect of access to credit on development outcomes (loan offer is determined by credit score threshold)
- Effect of wage increase on performance of mayors (Ferraz and Finan 2011, Gagliarducci and Nannicini 2013)
- Effect of wining office on wealth of politicians (Eggers and Hainmueller 2009)
- Effect of school district boundaries on home values (Black 1999)
- RD that exploits “close" elections has become a workhorse model for causal inference in electoral research
  - ...Lee, Moretti and Butler 2004, DiNardo and Lee 2004, Hainmueller and Kern 2008, Leigh 2008, Pettersson-Lidbom 2008, Broockman 2009, Butler 2009, Dal Bó, Dal Bó and Snyder 2009, Eggers and Hainmueller 2009, Ferreira and Gyourko 2009, Uppal 2009, 2010, Cellini, Ferreira and Rothstein 2010, Gerber and Hopkins 2011, Trounstine 2011, Boas and Hidalgo 2011, Folke and Snyder Jr. 2012, and Gagliarducci and Paserman 2012...
- many more...

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Estimating $\alpha_{SRDD} = \mathbb{E}[Y_{1i}|X_i = c] - \mathbb{E}[Y_{0i}|X_i = c]$

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3. Decide on a model for $\mathbb{E}[Y|X]$ on each side
   - linear, same slope for $\mathbb{E}[Y_0|X]$ and $\mathbb{E}[Y_1|X]$
   - linear, but different slopes
   - non-linear
   - good to start with visual inspection (scatter plot with kernel/lowess) to check which model is appropriate

SRRD Estimation 1: Linear with Same Slope

- $\mathbb{E}[Y_{0i}|X]$ is linear and treatment effect, α, does not depend on X_i:

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- Therefore $\mathbb{E}[Y_{1i}|X_i] = \alpha + \mathbb{E}[Y_{0i}|X_i] = \alpha + \mu + \beta X_i$
- Since D_i is determined given X_i, we have that:

$$\begin{aligned}
\mathbb{E}[Y_i|X_i, D_i] &= D \cdot \mathbb{E}[Y_{1i}|X_i] + (1 - D_i) \cdot \mathbb{E}[Y_{0i}|X_i] \\
&= \mu + \alpha D_i + \beta X_i \\
&= (\mu + \beta c) + \alpha D_i + \beta(X_i - c) \\
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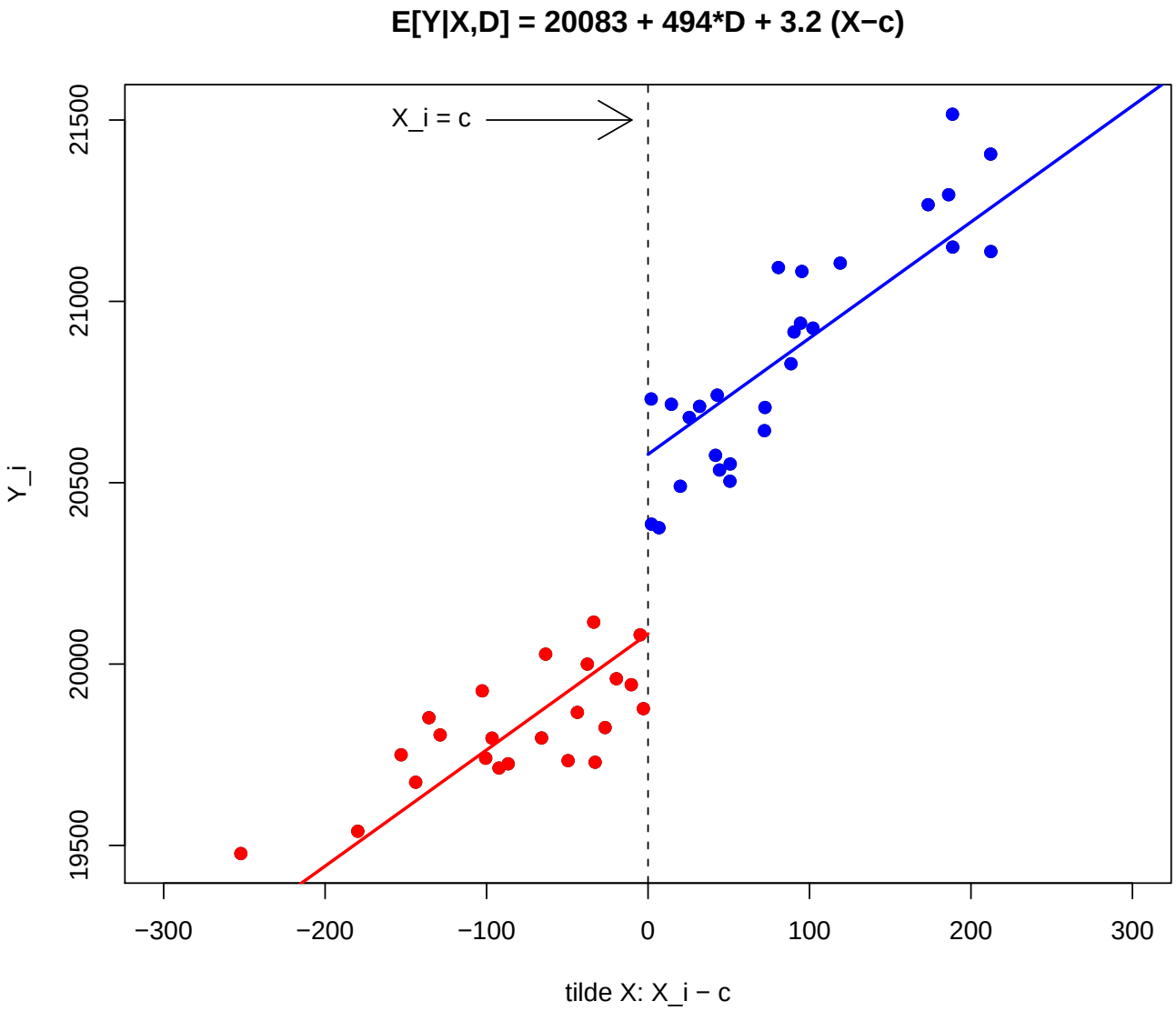
- So we just run a regression of Y on D and $\tilde{X}$

SRDD: Linear with Same Slope



Sharp RDD Estimation 2: Differential Slopes

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- Regress Y_i on $\tilde{X}_i$, D_i, and the interaction $\tilde{X}_i \cdot D_i$.

Sharp RDD Estimation 2: Differential Slopes

- Or, let $\mathbb{E}[Y_0|X]$ and $E[Y_1|X]$ be different linear functions of X, so the average effect of the treatment $E[Y_1 - Y_0|X]$ varies with X:

$$\mathbb{E}[Y_{0i}|X_i] = \mu_0 + \beta_0 X_i, \qquad E[Y_{1i}|X_i] = \mu_1 + \beta_1 X_i$$

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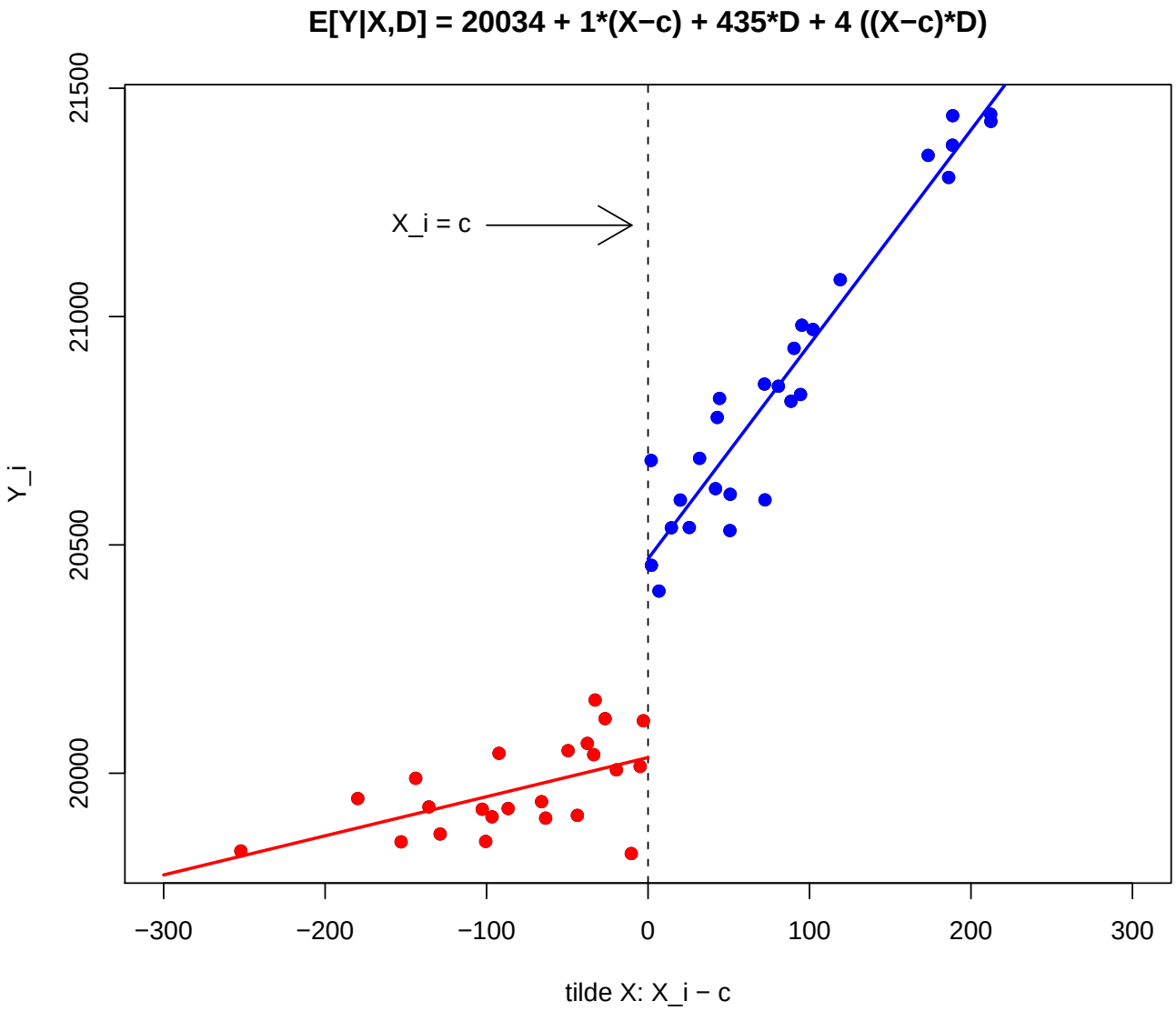
- What does each coefficient in the above represent?

SRDD: Linear with Differential Slope



Sharp RDD Estimation 3: Non-Linear Case

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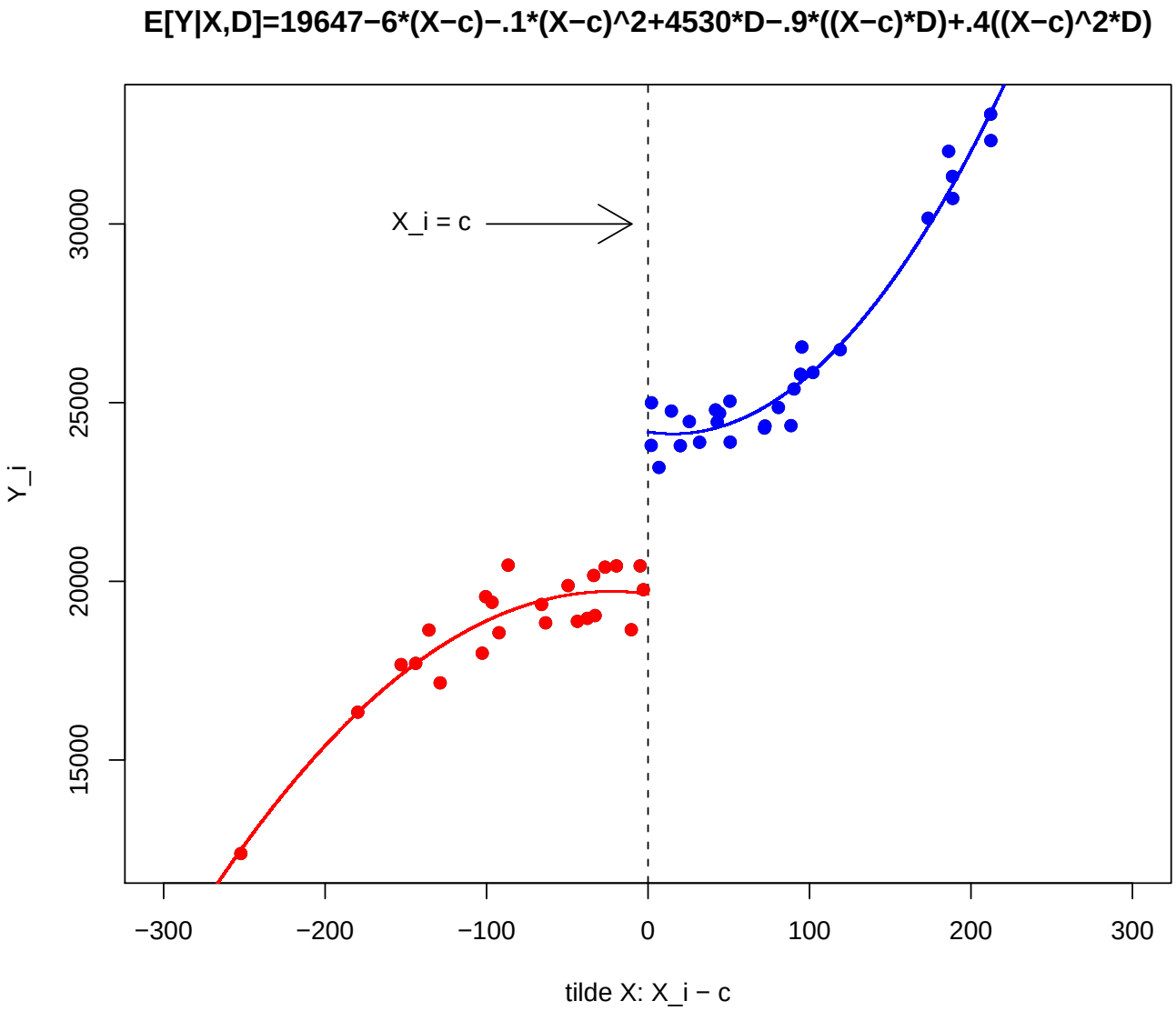
These models differ only in how we borrow information from neighboring areas of X to guess $\mathbb{E}[Y|X = c]$ from each side.

SRDD: Non-Linear Case



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- Also see Calonico, Cattaneo, and Titiunik for automatic bandwidth selection for local polynomial models, and automatic algorithms for a range of plots (see `rdrobust`)

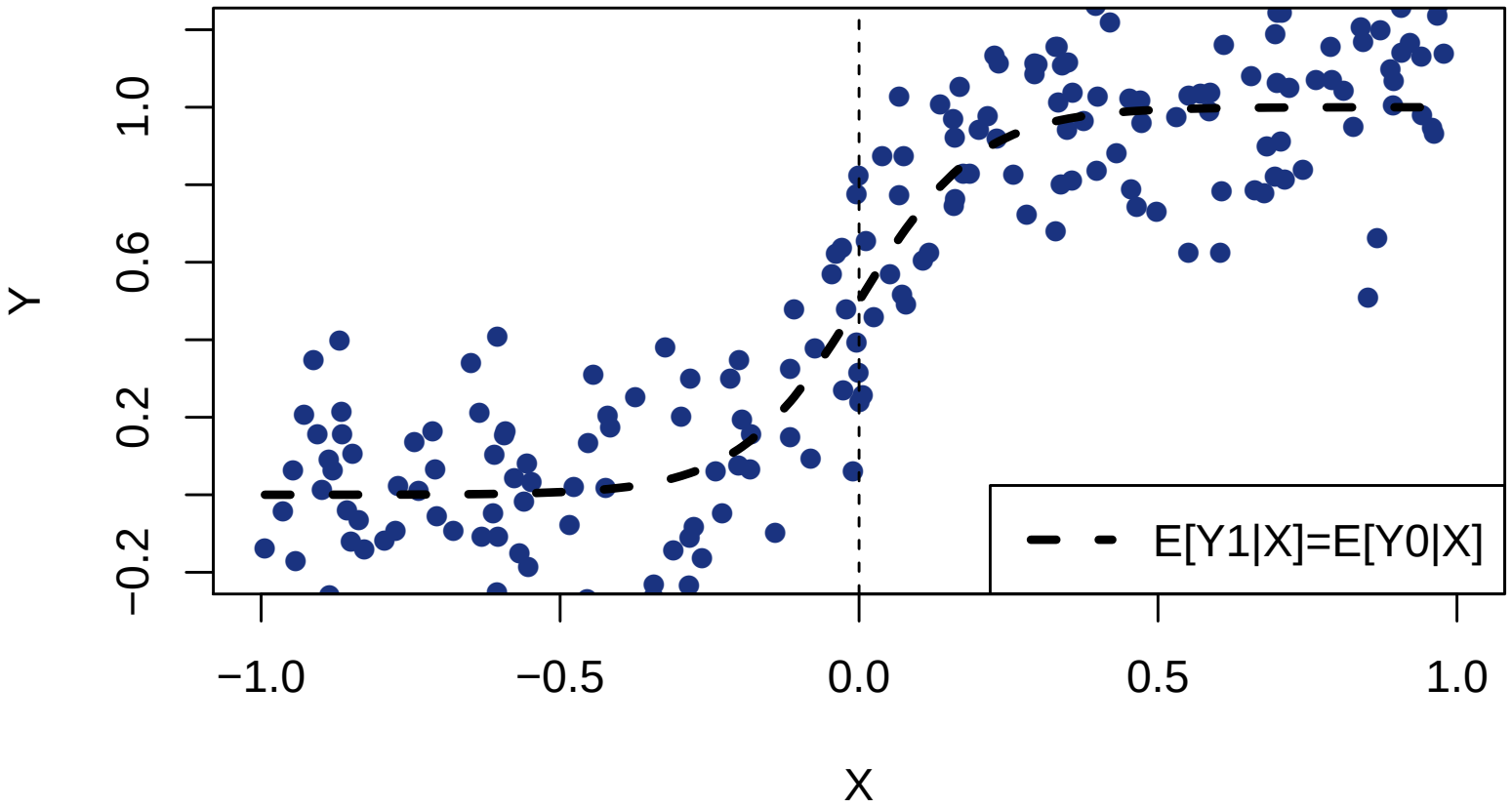
Outline

1. Sharp Regression Discontinuity Design
   - Identification
   - Sharp RRD Examples
   - Sharp RDD: Estimation
   - Falsification Checks
2. Fuzzy Regression Discontinuity Design
   - Identification
   - Estimation
   - Fuzzy RDD Example: The Effect of Early Release from Prison Marie (2008)

Sharp RDD: Falsification Checks

1. Sensitivity: Are results sensitive to alternative specifications?
2. Balance Checks: Do covariates Z jump at the threshold?
3. Check if jumps occur at placebo thresholds $c^{\star}$?
4. Sorting: Do units sort around the threshold?

Sharp RDD: Risk of Mispecification



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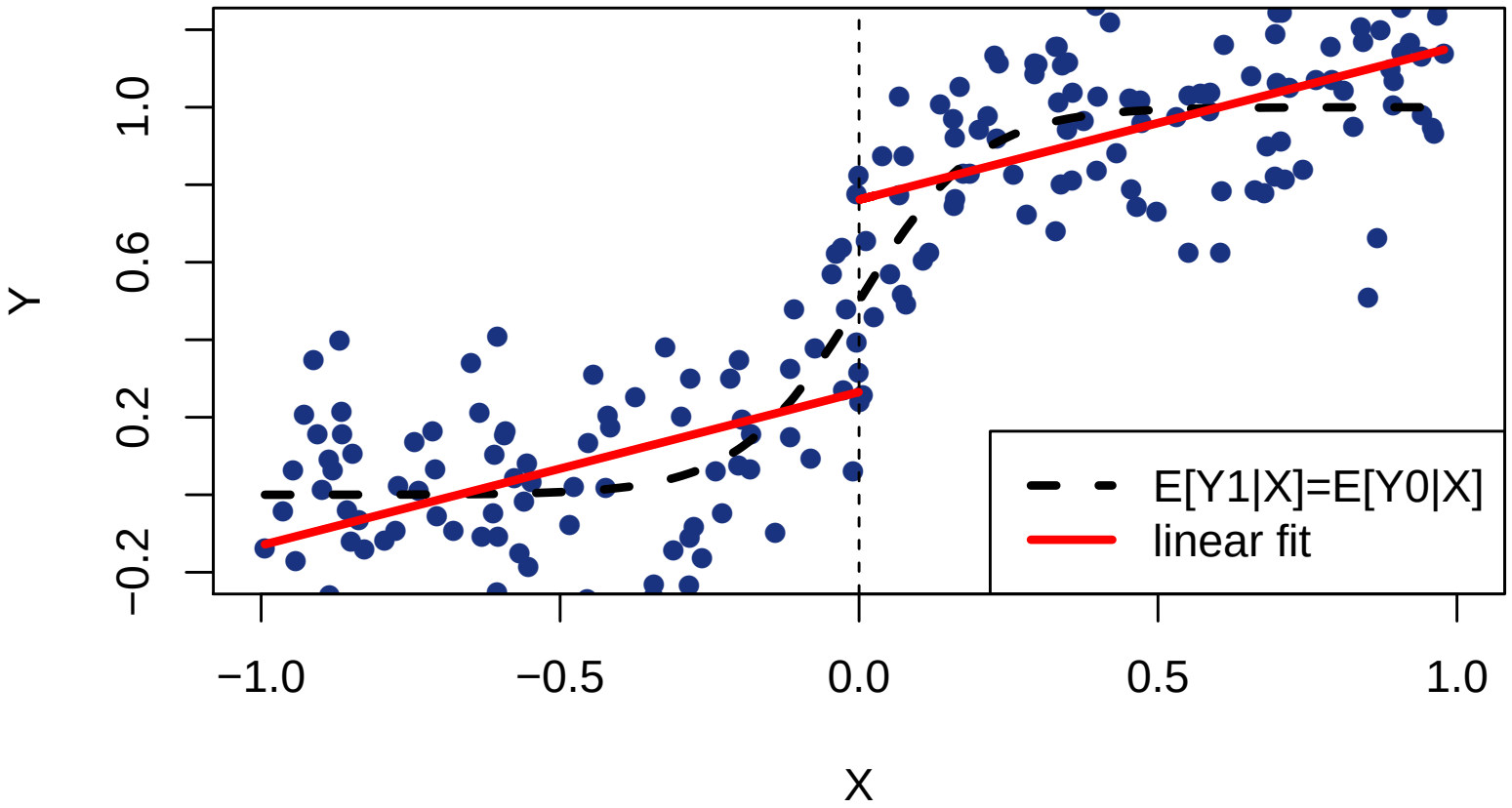

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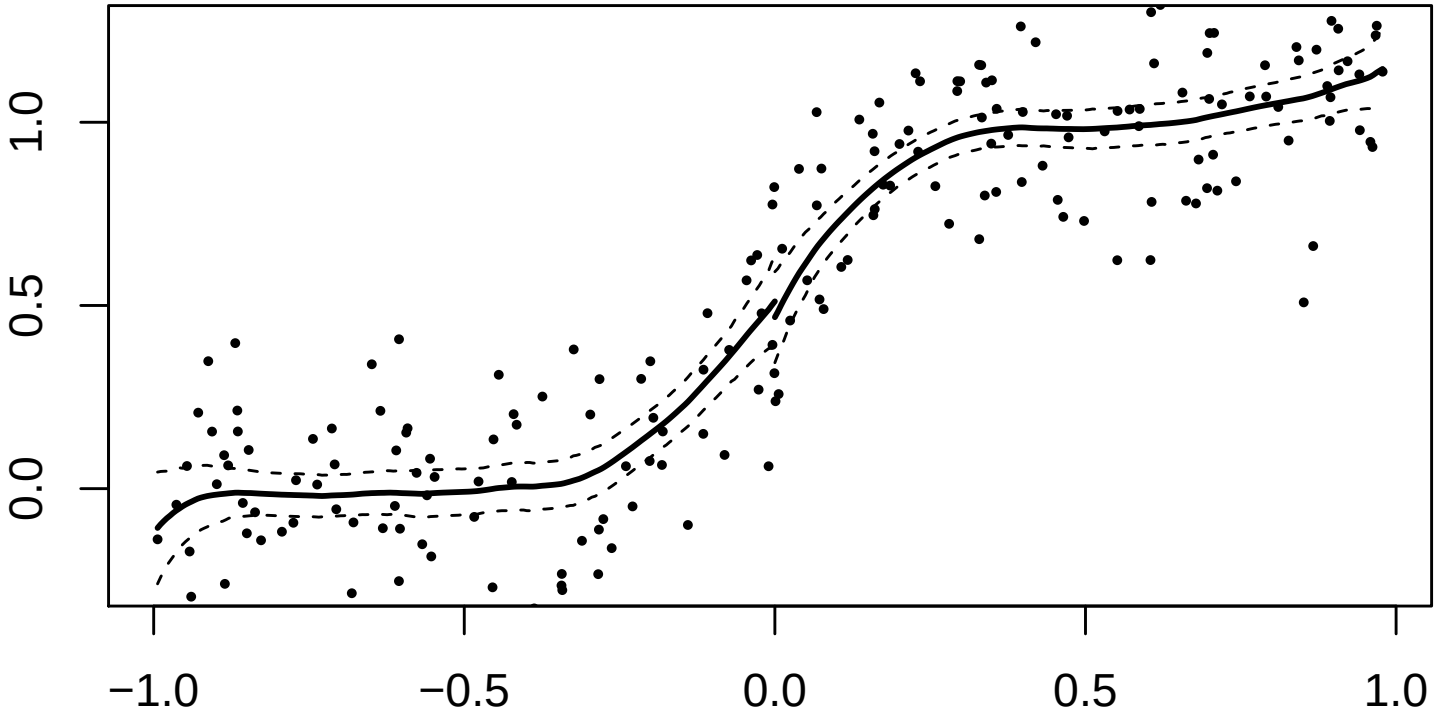
Using Imbens-Kalyanaraman bandwidth in `RDD`



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As usual, balance checks address only observables, not unobservables.

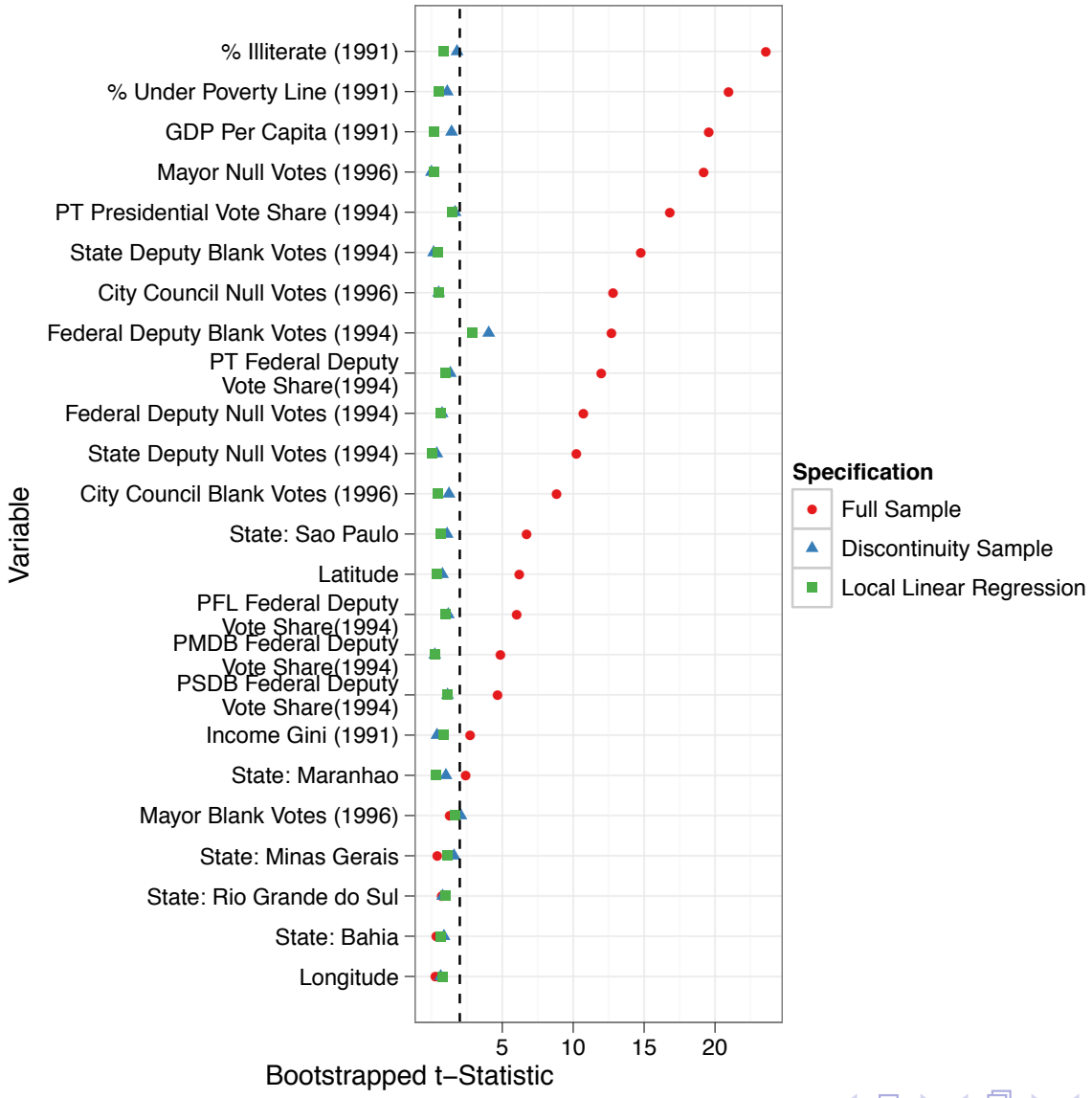
SRDD: You Can Add Covariates

TABLE 4. Regression Discontinuity Design Results: Effect of Serving in House of Commons on (Log) Wealth at Death

	Conservative Party		Labour Party	
Effect of serving	0.61	0.66	−0.20	−0.25
Standard error	(0.27)	(0.37)	(0.26)	(.26)
Covariates		x		x
Percent wealth increase	83	94	−18	−23
95% Lower bound	8	−7	−52	−65
95% Upper bound	212	306	31	71

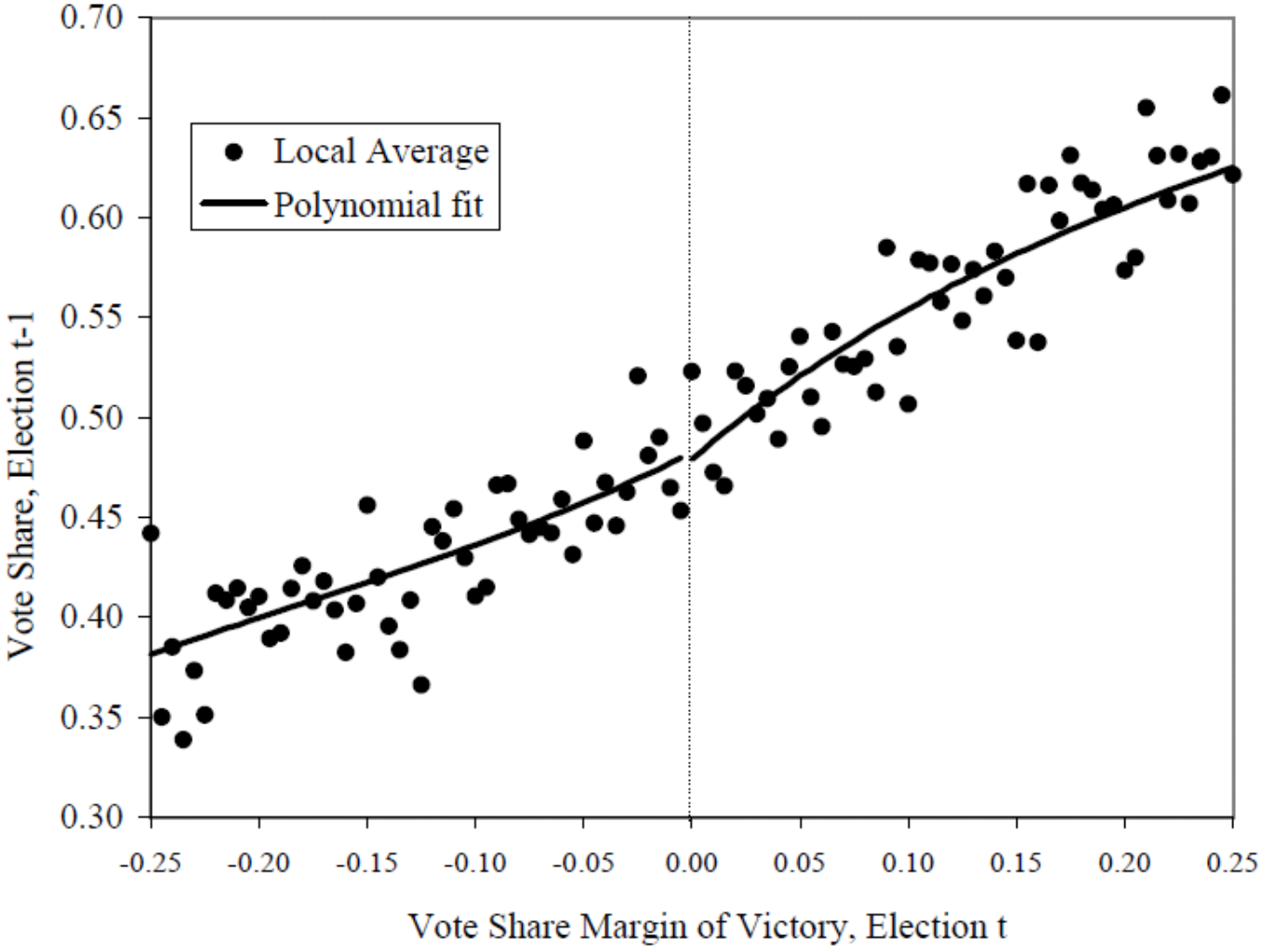
Note: Effect estimates at the threshold of winning $\tau_{RDD} = E[Y(1) - Y(0) \mid Z = 0]$. Estimates without covariates from local polynomial regression fit to both sides of the threshold with bootstrapped standard errors. Estimates with covariates from local linear regression with rectangular kernel (equation 2); bandwidth is 15 percentage point of vote share margin with

SRDD: Falsification Test



SRDD: Falsification Test

Especially convincing to do this with placebo outcome, i.e. covariates likely related to potential outcomes but not influence by treatment.



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Let $c^\star$ be a placebo threshold value. Run models

$$\mathbb{E}[Y_i|X_i, D_i] = \beta_0 + \beta_1(X_i - c^\star) + \alpha_{fake}D_i + \beta_3(X_i - c^\star) \cdot D_i$$

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and check if α_{fake} is large and significant.

- Usually we run separate models below and above c, with bandwidth to exclude c, to avoid misspecification due to inclusion of the actual jump.
- Detecting a jump at placebo threshold requires an explanation.
  - e.g. if other treatments occurred near $c^\star$, it could be okay
  - but problematic if it implies unknown treatments or unobserved influences may account for effect at c
  - can also reveal specification problems, as your model could be guessing a discontinuous jump occurs where it was actually smooth

SRDD: Sorting and Bundling as Discontinuities

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Bundled Treatments. What else changes at c? Bundled treatments can invalidate continuity assumption.

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  - Construct equal-sized non-overlapping bins of the forcing variable s.t. no bin includes points both above and below c
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- Formal test: McCrary (2008)

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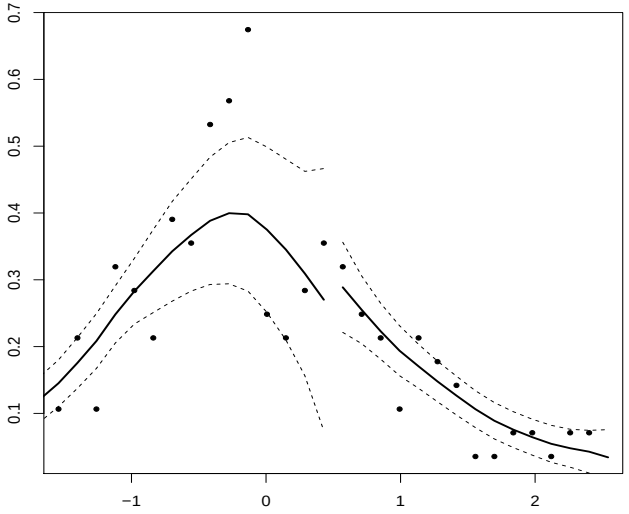

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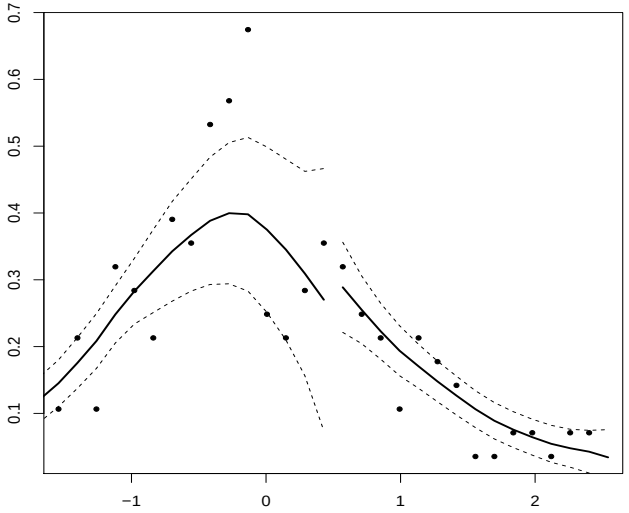


```
> sortprob=rep(0,N)
> sortprob[runvar<c]=.25*exp(runvar[runvar<c]-c)
> selfsorts=rbinom(N, size=1, prob = sortprob)
> runvar[selfsorts==1]=.5+
runif(sum(selfsorts),.001,.5)
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[1] 0.5857803
```
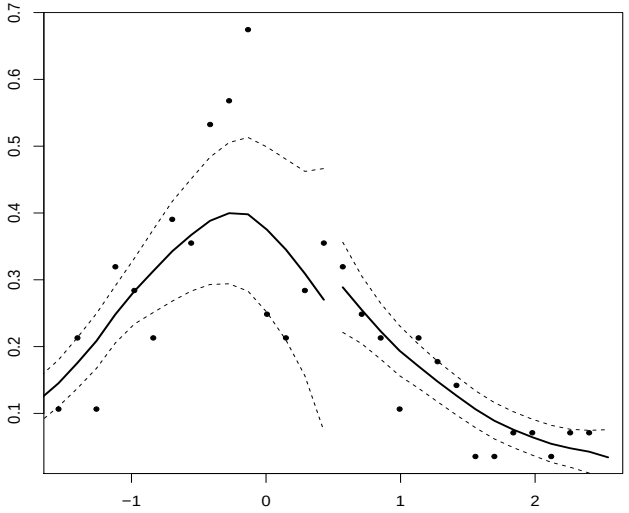
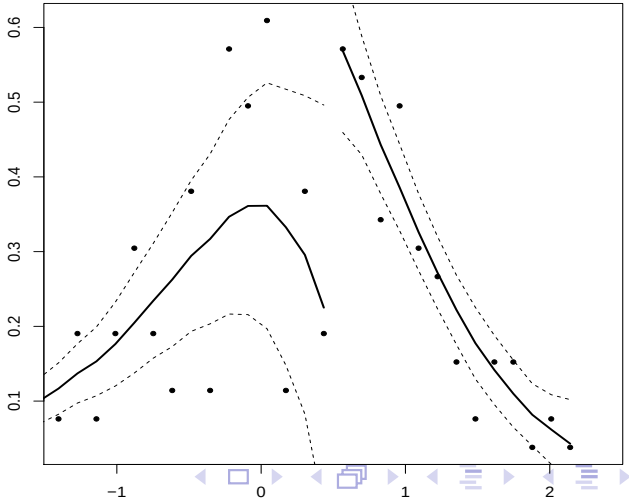




```
> sortprob=rep(0,N)
> sortprob[runvar<c]=.25*exp(runvar[runvar<c]-c)
> selfsorts=rbinom(N, size=1, prob = sortprob)
> runvar[selfsorts==1]=.5+
runif(sum(selfsorts),.001,.5)
> DCdensity(runvar, cutpoint=c)
[1] 0.0001952333
```



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1. Sharp Regression Discontinuity Design
   - Identification
   - Sharp RRD Examples
   - Sharp RDD: Estimation
   - Falsification Checks
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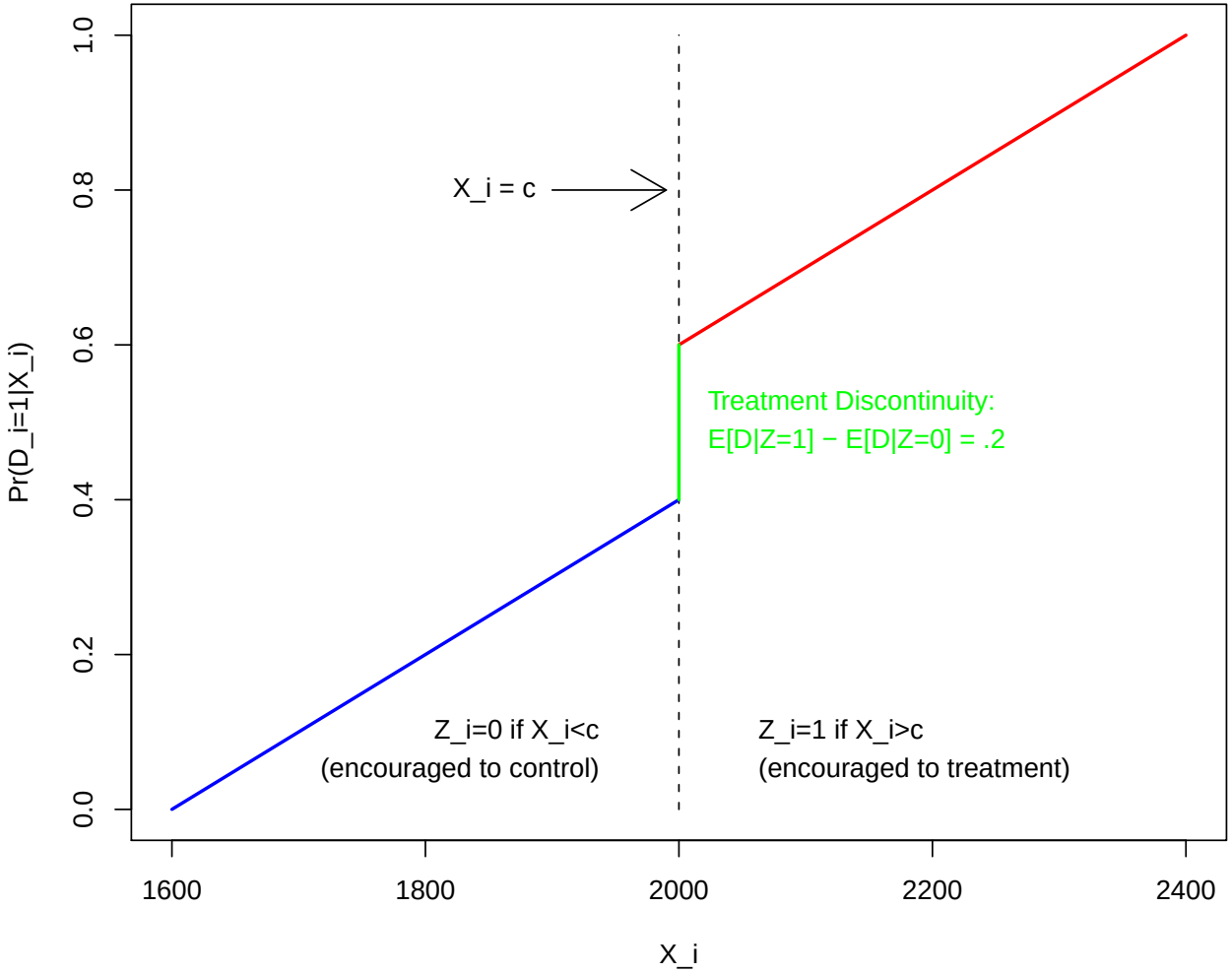
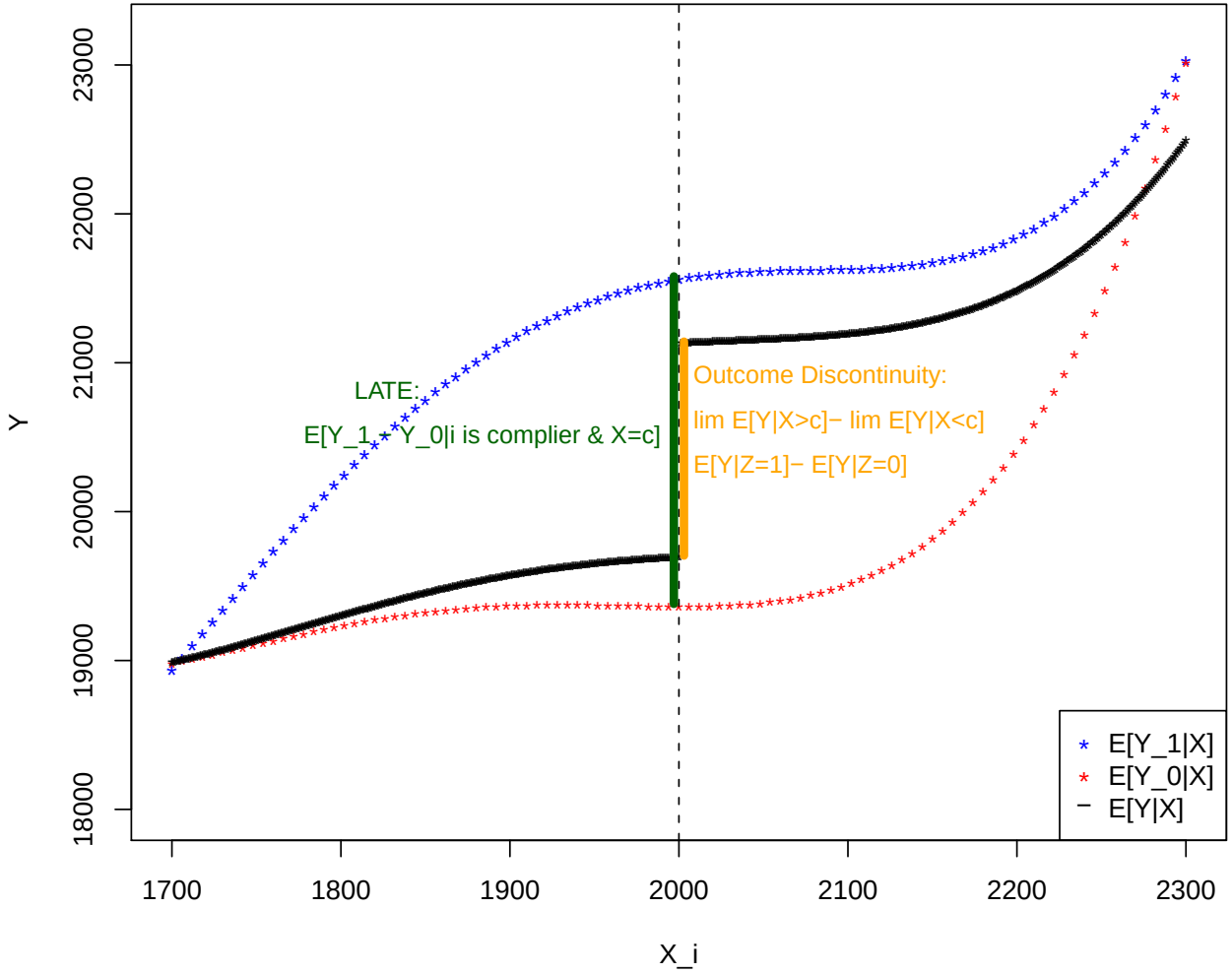
- Who are compliers in this framework?

Fuzzy RDD: Discontinuity in $\mathbb{E}[D|X]$



Fuzzy RDD: Discontinuity in $\mathbb{E}[Y|X]$



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Identification Result

$$\begin{aligned}
\alpha_{FRDD} &= \mathbb{E}[Y_{1i} - Y_{0i} | X_i = c \text{ and } i \text{ is a complier}] \\
&= \frac{\lim_{x \downarrow c} \mathbb{E}[Y_i | X_i = c] - \lim_{x \uparrow c} \mathbb{E}[Y_i | X_i = c]}{\lim_{x \downarrow c} \mathbb{E}[D_i | X_i = c] - \lim_{x \uparrow c} \mathbb{E}[D_i | X_i = c]} \\
&= \frac{\textit{outcome discontinuity}}{\textit{treatment discontinuity}} \\
&\approx \frac{\hat{\mathbb{E}}[Y | Z = 1, X = c] - \hat{\mathbb{E}}[Y | Z = 0, X = c]}{\hat{\mathbb{E}}[D | Z = 1, X = c] - \hat{\mathbb{E}}[D | Z = 0, X = c]}
\end{aligned}$$

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- Also good: separately estimate the outcome discontinuity and treatment discontinuity, then divide. Can use any estimator this way.

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- How valid are the identification assumptions?

Table 1: Descriptive Statistics for Prisoners Released by Length of Sentence and HDC and Non HDC Discharges

Panel A - Released Before 3 Months:			
Discharge Type	**Non HDC**	**HDC**	**Total**
Percentage Female	12.2	-	12.2
Mean Age	29.5	-	29.5
Percentage Incarcerated for Violence	17.6	-	17.6
Mean Number Previous Offences	8.8	-	8.8
Recidivism within 12 Months	52.4	-	52.4
Sample Size	42,987	0	42,987
Panel B - Released Between 3 and 6 Months:			
Discharge Type	**Non HDC**	**HDC**	**Total**
Percentage Female	8.8	8.8	8.8
Mean Age at Release	27.6	30.8	28.4
Percentage Incarcerated for Violence	20.3	18.3	19.8
Mean Number Previous Offences	10	6.5	9.1
Recidivism within 12 Months	60	30.2	52.6
Sample Size	52,091	17,222	69,313

Table 2: Descriptive Statistics for Prisoners Released by Length of Sentence and HDC and Non HDC Discharges and +/-7 Days Around Discontinuity Threshold

Panel A - Released +/- 7 Days of 3 Months (88 Days) Cut-off:			
Discharge Type	**Non HDC**	**HDC**	**Total**
Percentage Female	10.5	9.7	10.3
Mean Age at Release	28.9	30.7	29.3
Percentage Incarcerated for Violence	19.8	18.2	19.4
Mean Number Previous Offences	9.5	5.7	8.7
Recidivism within 12 Months	54.6	28.1	48.8
Sample Size	18,928	5,351	24,279
Panel B - Released +/- 7 Days of 3 Months (88 Days) Cu-off:			
Day of Release around Cut-off	**- 7 Days**	**+ 7 Days**	**Total**
Percentage Female	11	10.2	10.3
Mean Age at Release	28.8	29.4	29.3
Percentage Incarcerated for Violence	17.1	19.7	19.4
Mean Number Previous Offences	9.1	8.6	8.7
Recidivism within 12 Months	56.8	47.9	48.8
Percentage Released on HDC	0	24.4	22
Sample Size	2,333	21,946	24,279

Figure 1: Proportion Discharged on HDC by Sentence Length

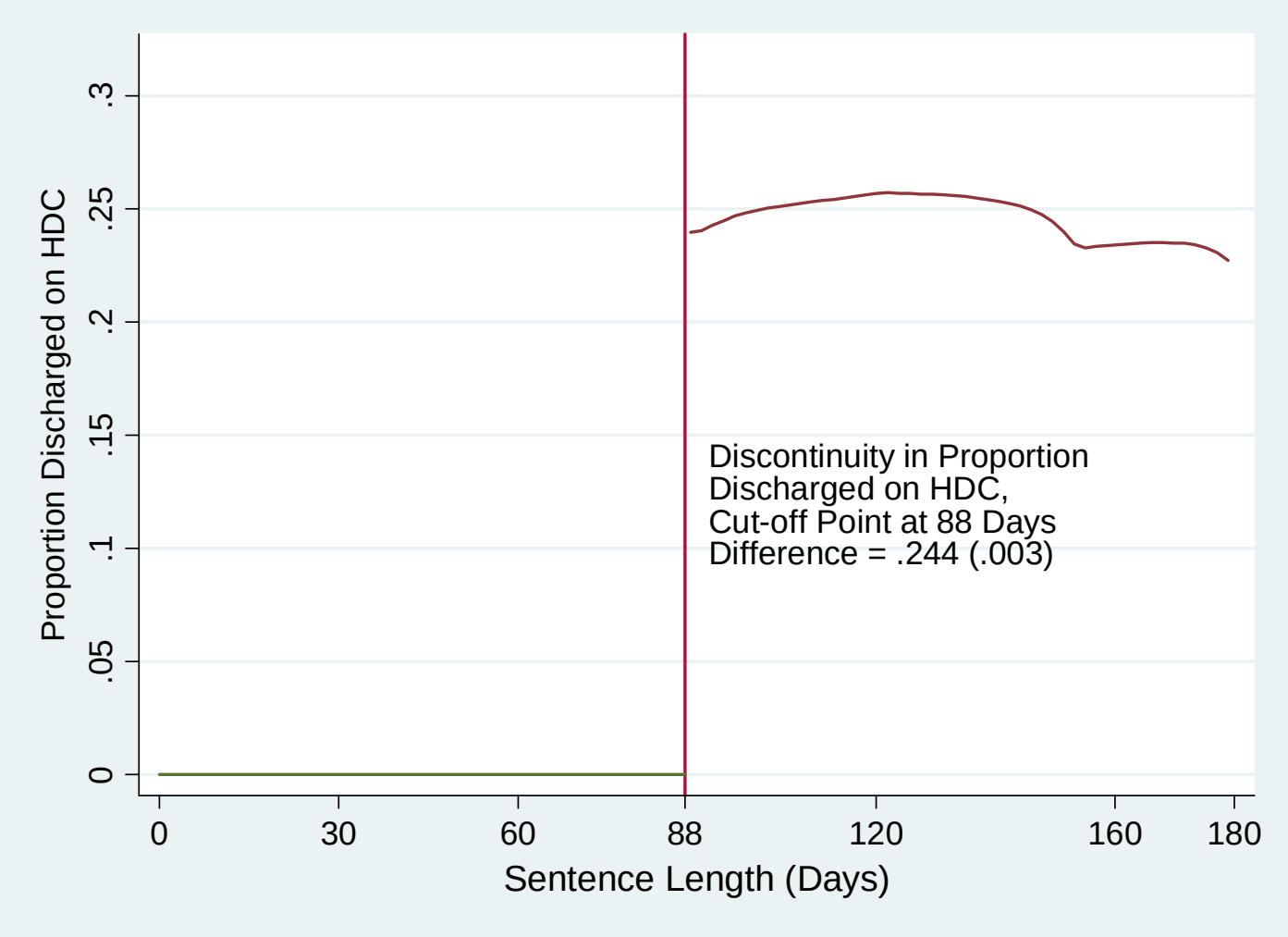

Figure 2: Mean Number of Previous Offence by Sentence Length

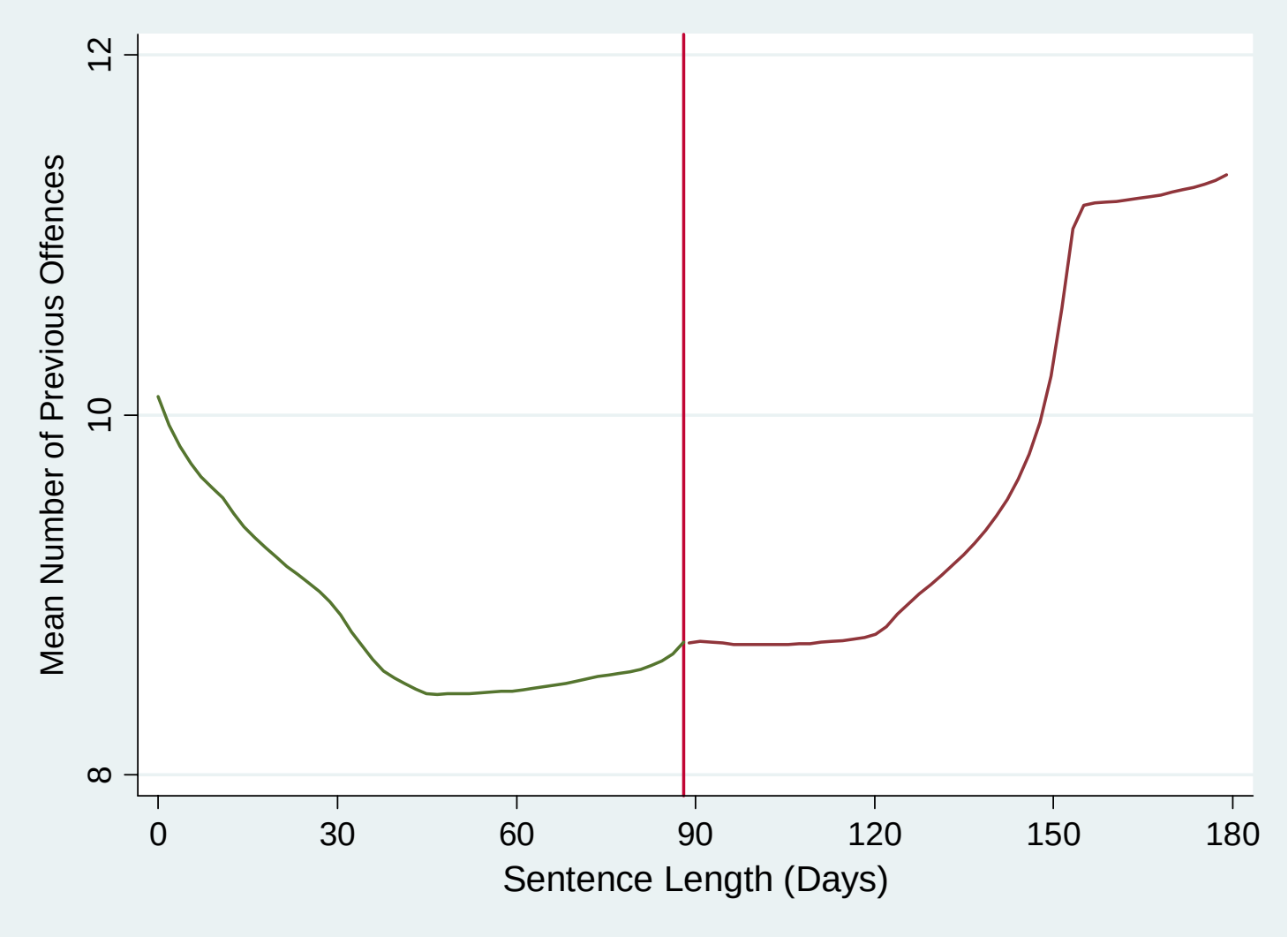

Table 4: RDD Estimates of HDC Impact on Recidivism – Around Threshold

	Dependent Variable = Recidivism Within 12 Months Estimation on Individuals Discharged +/- 7 Days of 88 Days Threshold		
	(1)	(2)	(3)
Estimated Discontinuity of HDC Participation at Threshold ($HDC^{+} - HDC^{-}$)	.243 (.009)	.223 (.009)	.243 (.003)
Estimated Difference in Recidivism Around Threshold ($Rec^{+} - Rec^{-}$)	-.089 (.011)	-.059 (.009)	-.044 (.014)
Estimated Effect of HDC on Recidivism Participation ($Rec^{+} - Rec^{-}$)/ ($HDC^{+} - HDC^{-}$)	-.366 (.044)	-.268 (.044)	-.181 (n.a.)
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PSM	No	No	Yes
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For what group is this the ATE?

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- Only with strong assumptions (e.g., homogenous treatment effects) can we estimate the overall average treatment effect
- So, RDDs have strong internal validity but may have weak external validity